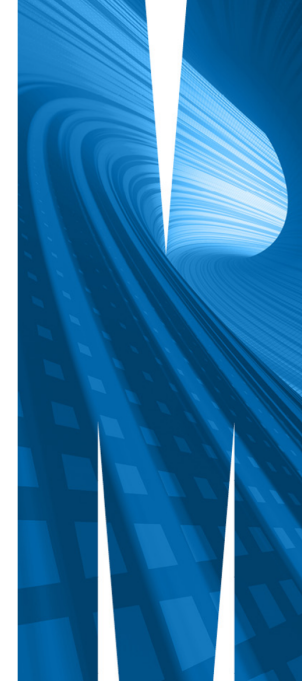
MONASH University

ETX1100/ETX5900 Business Statistics

Week 4: Testing business claims

15 December 2026



Outline

1. From claims to hypotheses
2. Tests for a population mean
3. Tests for a population proportion
4. Decisions, errors, and importance
5. Integrated evidence brief

Today’s journey

1. Translate a business claim into hypotheses.
2. Use one five-step testing process.
3. Test population means and proportions in Excel.
4. Interpret p-values, decisions, and possible errors.
5. Separate statistical evidence from practical importance.

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Retrieval check

- What does a confidence interval estimate?
- What is a standard error?
- When do we use *t* rather than *z* for a mean?
- Why must hypotheses refer to population parameters rather than sample statistics?

What hypothesis testing does

Hypothesis testing asks whether sample evidence is difficult to reconcile with a stated population benchmark.

- Begin with a business claim.
- Assume a null benchmark temporarily.
- Measure how far the sample result is from that benchmark.
- Quantify extremeness with a p-value.
- Make a decision with a pre-selected significance level.

The five-step workflow

1. State H_0 and H_1.
2. Calculate the test statistic.
3. Calculate the p-value in the correct tail or tails.
4. Compare the p-value with α.
5. State a contextual conclusion about H_1.

Null and alternative hypotheses

- H_0 contains equality and represents the benchmark.
- H_1 expresses the claim being investigated.
- Right-sided: $H_1 : \theta > \theta_0$.
- Left-sided: $H_1 : \theta < \theta_0$.
- Two-sided: $H_1 : \theta \neq \theta_0$.

Choose direction from the question before seeing the result.

Watch me: translate a click-through claim

An analyst claims the population mean click-through rate is below 12 clicks per minute.

$$H_0 : \mu = 12, \qquad H_1 : \mu < 12.$$

The parameter is μ, not the observed $\bar{x}$. The word **below** makes this a left-sided test.

Now you try: write the hypotheses

Time allowed: 5 minutes

Write hypotheses and identify the tail for each claim:

1. Average delivery time exceeds 30 minutes.
2. Mean transaction value differs from AUD 85.
3. Fewer than 4% of invoices contain an error.
4. More than half of customers use the loyalty app.

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Test statistic for a mean

When σ is unknown:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1.$$

The statistic is the observed difference measured in standard errors. Its sign indicates direction; its magnitude indicates distance from H_0.

P-values and tails in Excel

Alternative	Excel p-value
$H_1 : \mu < \mu_0$	`T.DIST(t,df,TRUE)`
$H_1 : \mu > \mu_0$	`1-T.DIST(t,df,TRUE)`
$H_1 : \mu \neq \mu_0$	`T.DIST.2T(ABS(t),df)`

Using `ABS` with `T.DIST.2T` avoids separate positive/negative formulas.

What a p-value means

The p-value is the probability—assuming H_0 is true—of obtaining a test statistic at least as extreme as the one observed, in the direction specified by H_1.

It is not:

- the probability that H_0 is true;
- the size or importance of an effect; or
- the probability that the result occurred “by chance”.

Watch me: test click-through rate

Given $n = 45$, $\bar{x} = 10.8$, $s = 3.2$, $\mu_0 = 12$, and $\alpha = 0.05$:

1. $t = (10.8 - 12)/(3.2/\sqrt{45})$;
2. use `=T.DIST(t,44,TRUE)`;
3. compare the p-value with 0.05; and
4. conclude about the population mean click-through rate.

Now you try: discretionary spending

Time allowed: 8 minutes

A random sample of 40 students has $\bar{x} = 283$ and $s = 17.85$. Test at $\alpha = 0.05$ whether population mean monthly discretionary spending exceeds AUD 275.

Complete all five steps, show the Excel p-value formula, and state the conclusion without saying “accept H_0”.

One-sided versus two-sided

A two-sided test considers evidence in both directions and uses both tails. A one-sided test concentrates the rejection region in one pre-specified direction.

Do not choose one-sided after observing the sample. That inflates the chance of finding a convenient result.

Now you try: change only the claim

Time allowed: 6 minutes

Using the same spending sample, test whether the mean **differs from** AUD 275.

1. Which steps change?
2. Does the test statistic change?
3. Which p-value formula changes?
4. Compare the one- and two-sided conclusions.

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Large-sample proportion test

For $H_0 : \pi = \pi_0$:

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\pi_0(1 - \pi_0)/n}}.$$

Check:

$$n\pi_0 \geq 5, \qquad n(1 - \pi_0) \geq 5.$$

The null value π_0 appears in the test standard error.

Proportion p-values in Excel

Alternative	Excel p-value
$H_1 : \pi < \pi_0$	`NORM.S.DIST(z,TRUE)`
$H_1 : \pi > \pi_0$	`1-NORM.S.DIST(z,TRUE)`
$H_1 : \pi \neq \pi_0$	`2*(1-NORM.S.DIST(ABS(z),TRUE))`

The tail follows the alternative hypothesis, not merely the sign of z.

Watch me: transaction success rate

A bank benchmarks error-free processing at 95%. In a random sample of 250 transactions, 229 succeed.

1. $\hat{p} = 229/250 = 0.916$.
2. Test $H_0 : \pi = 0.95$ against $H_1 : \pi < 0.95$.
3. Verify the large-count condition using π_0.
4. Calculate z and its left-tail p-value.
5. Conclude at $\alpha = 0.10$.

Now you try: invoice accuracy

Time allowed: 8 minutes

A company claims at least 98% of invoices are accurate. A random audit finds 288 accurate invoices out of 300.

Test at $\alpha = 0.05$ whether the population accuracy rate is below 98%.

Include the assumption check, statistic, Excel p-value, decision, and business conclusion.

Confidence intervals and tests agree

For a two-sided test at significance α:

- reject $H_0 : \theta = \theta_0$ when the corresponding $(1 - \alpha)$ confidence interval excludes θ_0;
- fail to reject when it includes θ_0.

The interval also communicates plausible effect sizes, which the p-value alone does not.

Now you try: interval cross-check

Time allowed: 6 minutes

For the invoice sample:

1. construct a 95% confidence interval for π;
2. compare it with the 98% benchmark;
3. explain why this is a two-sided cross-check, while the original claim was left-sided; and
4. identify what the interval adds to the decision.

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Decision language

If p-value $< \alpha$:

Reject H_0; there is sufficient evidence that ...

If p-value $\geq \alpha$:

Fail to reject H_0; there is insufficient evidence that ...

"Fail to reject" is not proof that the null is true.

Type I and Type II errors

Decision	H_0 true	H_0 false
Reject H_0	Type I error	correct
Fail to reject H_0	correct	Type II error

- P(Type I error) = α.
- Reducing α makes rejection harder and can increase Type II risk if other factors stay fixed.

Watch me: errors in context

For $H_1 : \mu > 275$:

- **Type I:** conclude mean spending exceeds AUD 275 when it does not.
- **Type II:** fail to find evidence that mean spending exceeds AUD 275 when it actually does.

Name the real-world cost of each error before choosing α.

Now you try: business consequences

Time allowed: 6 minutes

For the invoice-accuracy test:

1. describe Type I and Type II errors in context;
2. identify who bears the cost of each;
3. recommend whether 10%, 5%, or 1% significance is most defensible; and
4. justify the recommendation without calculating a new test.

Statistical versus practical significance

- Statistical significance asks whether the result is difficult to explain by sampling variation under H_0.
- Practical significance asks whether the effect is large enough to matter.
- Large samples can make tiny effects statistically significant.
- Report an effect estimate and confidence interval alongside the p-value.

Now you try: challenge the headline

Time allowed: 5 minutes

A report says, “Waiting time fell by 0.2 minutes, $p < 0.001$, so the redesign was a major success.”

Identify:

1. what the p-value supports;
2. what it does not support;
3. additional quantities needed; and
4. a more defensible conclusion.

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Watch me: audit a testing worksheet

Check that the spreadsheet separates:

- inputs and units;
- hypotheses and chosen α;
- assumption checks;
- statistic and p-value;
- decision; and
- contextual conclusion.

Trace formulas to ensure the sample statistic, benchmark, and tail are consistent.

Now you try: complete the evidence brief

Time allowed: 10 minutes

Choose either a mean claim from `week4-testing-data.xlsx` or a product-trial proportion from `week4-proportions-data.xlsx`.

Produce a one-page brief containing the five test steps, an interval or effect estimate, one possible error, and a two-sentence recommendation.

Common traps

- Writing hypotheses with $\bar{x}$ or $\hat{p}$.
- Putting equality in H_1.
- Choosing the tail after seeing the data.
- Using $\hat{p}$ rather than π_0 in a test standard error.
- Saying a p-value is the probability that H_0 is true.
- Saying “accept H_0”.
- Treating statistical significance as business importance.

Closing check

Can you:

- translate a claim into directional hypotheses?
- complete the five-step process for a mean or proportion?
- select the correct Excel tail?
- state a decision and conclusion precisely?
- describe Type I and Type II errors in context?
- distinguish evidence strength from effect importance?